

WASP-50b

R-0778

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_221123 · created 2026-07-28T05:52:01+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459906.763 +/- 0.081 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.156 +/- 0.048
Transit depth (Rp/Rs)^2	2.44 +/- 1.51 [%]
Orbital Inclination (inc)	82.5 +/- 2.81
Airmass coefficient 1 (a1)	1.013 +/- 0.017
Airmass coefficient 2 (a2)	-0.008 +/- 0.031
Scatter in the residuals of the lightcurve fit is	1.68 %
Best Comparison Star	#2 - [362, 303]
Optimal Method	PSF photometry
Transit Duration (day)	0.028 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.156 ± 0.048 vs 0.1404 (deviation 0.33σ)
Duration fit vs published	0.028 d vs 0.0754 d (deviation 1.53σ)
Mid-time O–C	34.2 min
Depth SNR	3.2
Residual scatter	1.68 %

Light curve

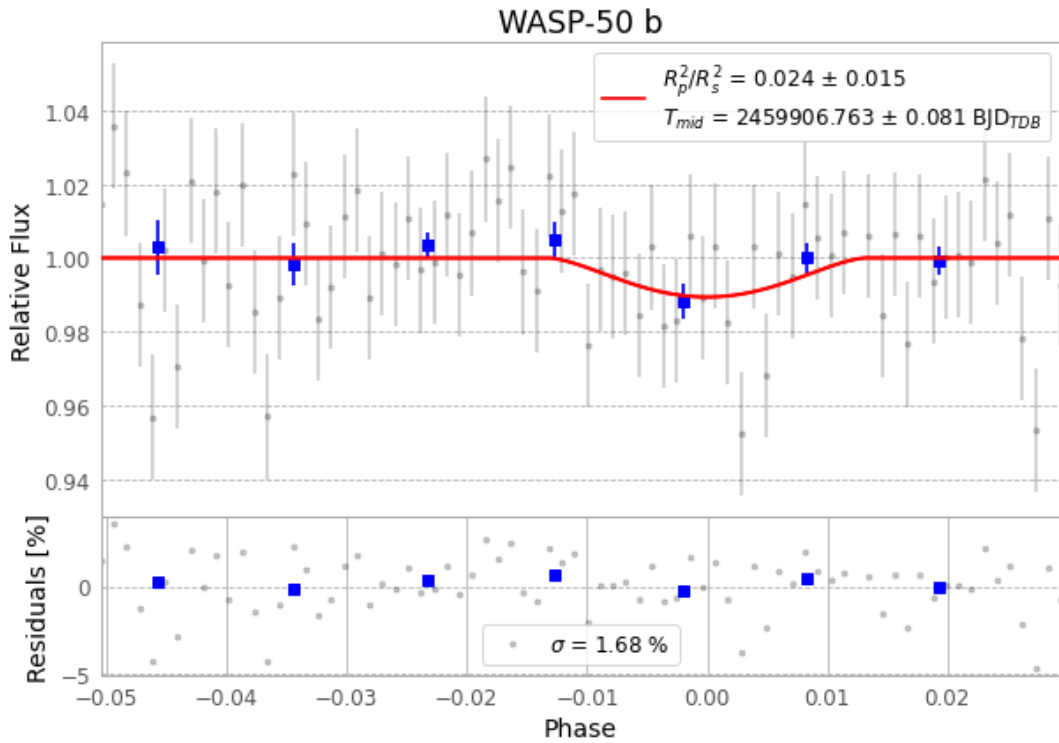


Figure 1 — FinalLightCurve_WASP-50 b_2022-11-22.png (SHA-256 d359c15c2f59b8e8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [310, 290], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 db68bf99d7c373c7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

76 FITS frames (50 MB), WASP-50221123034815.FITS → WASP-50221123073315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-221123.log (7c152009ba46...), FinalLightCurve_WASP-50 b_2022-11-22.png (d359c15c2f59...), FinalLightCurve_WASP-50 b_2022-11-22.csv (823faee07992...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 05:52Z.